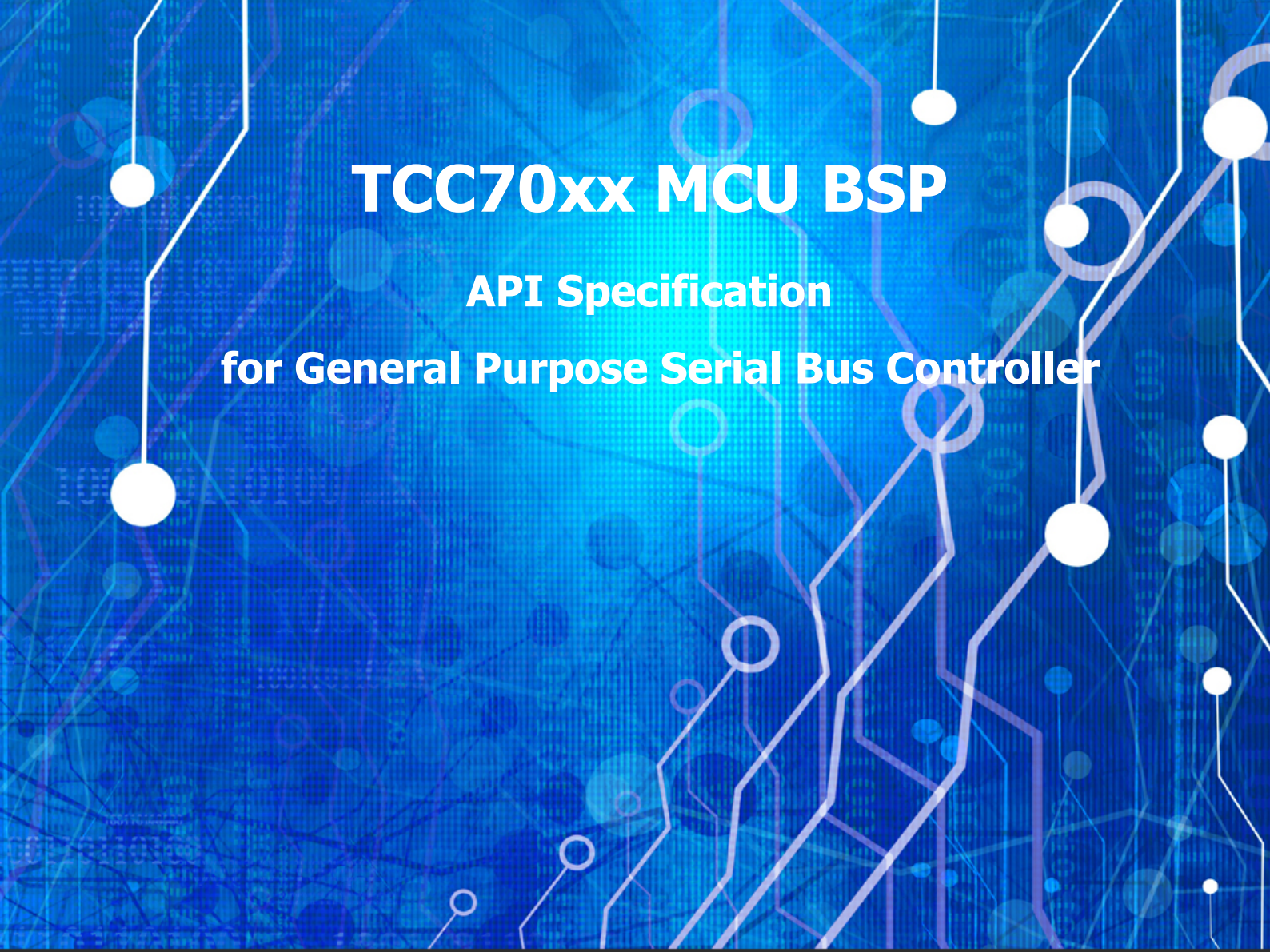




TCC70xx MCU BSP

API Specification

for General Purpose Serial Bus Controller

Rev. 0.10 [A]

2021-09-03

Preliminary version

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1 INTRODUCTION

This document describes the general purpose serial bus controller device driver (hereinafter referred to as GPSB driver) of TCC70xx MCU BSP. For more detailed information on general purpose serial bus controller in the TCC70xx MCU Sub-system, refer to Chapter 9 General Purpose Serial Bus (GPSB) Controller in "*TCC70xx Full Specification*". [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

GPSB	General Purpose Serial Bus
BPW	Bit Per Width
SCK	GPSB (SPI) Clock
SDO	GPSB (SPI) Output (Master channel output, Slave channel is <u>not</u> supported.)
SDI	GPSB (SPI) Input (Master channel input, Slave channel is <u>not</u> supported.)
CS	Chip Select PIN of GPSB

3 SOURCE FILES

3.1 Location of Source Files

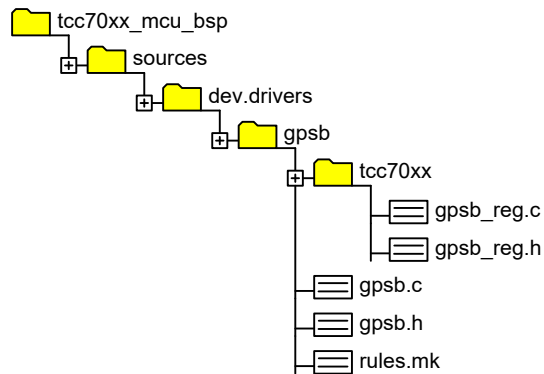


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- `gpsb_reg.c`
 - GPSB specific source code
- `gpsb_reg.h`
 - GPSB specific header
- `gpsb.c`
 - GPSB common source code
- `gpsb.h`
 - GPSB common header including API & Defines
- `rules.mk`
 - Rules used for build

4 CONSTANTS

4.1 GPSB Base Address

Constant values for base register address for each GPSB channel and GPSB configuration. They are used to change register value for the GPSB operation.

Declaration

```
#define GPSB0_BASE      (MCU_BSP_GPSB_BASE + 0x00000)
#define GPSB1_BASE      (MCU_BSP_GPSB_BASE + 0x10000)
#define GPSB2_BASE      (MCU_BSP_GPSB_BASE + 0x20000)
#define GPSB3_BASE      (MCU_BSP_GPSB_BASE + 0x30000)
#define GPSB4_BASE      (MCU_BSP_GPSB_BASE + 0x40000)
#define GPSB5_BASE      (MCU_BSP_GPSB_BASE + 0x50000)
#define GPSB_PCFG_BASE   (MCU_BSP_GPSB_BASE + 0xF0000)
#define GPSB_IO_MONITOR_BASE (MCU_BSP_GPSB_BASE + 0xE0000)
```

Description

Value	Description
MCU_BSP_GPSB_BASE	GPSB controller register base address: 0xA0100000UL
GPSB0_BASE	GPSB controller channel 0 base address: 0xA0100000UL
GPSB1_BASE	GPSB controller channel 1 base address: 0xA0110000UL
GPSB2_BASE	GPSB controller channel 2 base address: 0xA0120000UL
GPSB3_BASE	GPSB controller channel 3 base address: 0xA0130000UL
GPSB4_BASE	GPSB controller channel 4 base address: 0xA0140000UL
GPSB5_BASE	GPSB controller channel 5 base address: 0xA0150000UL
GPSB_PCFG_BASE	GPSB controller port configuration base address: 0xA01F0000UL
GPSB_IO_MONITOR_BASE	GPSB controller IO monitor base address: 0xA01E0000UL

Remark

None

4.2 GPSB Channel

Constant values for index of each channel

Declaration

```
#define GPSB_CH_0    (0UL)
#define GPSB_CH_1    (1UL)
#define GPSB_CH_2    (2UL)
#define GPSB_CH_3    (3UL)
#define GPSB_CH_4    (4UL)
```

Description

Value	Description
<i>GPSB_CH_0</i>	<i>GPSB Channel 0 index value</i>
<i>GPSB_CH_1</i>	<i>GPSB Channel 1 index value</i>
<i>GPSB_CH_2</i>	<i>GPSB Channel 2 index value</i>
<i>GPSB_CH_3</i>	<i>GPSB Channel 3 index value</i>
<i>GPSB_CH_4</i>	<i>GPSB Channel 4 index value</i>

Remark

None

4.3 GPSB Mode

Constant values for the GPSB transfer mode as parameter of [GPSB_Xfer](#) function

Declaration

```
#define GPSB_XFER_MODE_WITHOUT_INTERRUPT    (uint32)(0x01UL)
#define GPSB_XFER_MODE_WITH_INTERRUPT      (uint32)(0x02UL)
#define GPSB_XFER_MODE_DMA                 (uint32)(0x10UL)
#define GPSB_XFER_MODE_PIO                 (uint32)(0x20UL)
#define GPSB_XFER_MODE_WITHOUT_CTF        (uint32)(0x40UL)
```

Description

Value	Description
<i><code>GPSB_XFER_MODE_WITHOUT_INTERRUPT</code></i>	<i>Non-interrupt transfer mode</i>
<i><code>GPSB_XFER_MODE_WITH_INTERRUPT</code></i>	<i>Interrupt transfer mode</i>
<i><code>GPSB_XFER_MODE_DMA</code></i>	<i>DMA transfer mode</i>
<i><code>GPSB_XFER_MODE_PIO</code></i>	<i>PIO (FIFO) transfer mode</i>
<i><code>GPSB_XFER_MODE_WITHOUT_CTF</code></i>	<i>Continuous transfer mode (Keep CS signal activate state)</i>

Remark

None

4.4 GPSB Event

Constant values for the GPSB error event. When a GPSB error interrupt occurs, the GPSB driver notifies you of the errors via a callback function.

Declaration

```
#define GPSB_EVENT_COMPLETE (uint32)(0x0000UL)
#define GPSB_EVENT_ERR_ROR (uint32)(0x0100UL)
#define GPSB_EVENT_ERR_WUR (uint32)(0x0200UL)
#define GPSB_EVENT_ERR_RUR (uint32)(0x0400UL)
#define GPSB_EVENT_ERR_WOR (uint32)(0x0800UL)
```

Description

Value	Description
<i>GPSB_EVENT_COMPLETE</i>	<i>Transfer complete</i>
<i>GPSB_EVENT_ERR_ROR</i>	<i>Read FIFO over-run error</i>
<i>GPSB_EVENT_ERR_WUR</i>	<i>Write FIFO under-run error</i>
<i>GPSB_EVENT_ERR_RUR</i>	<i>Read FIFO under-run error</i>
<i>GPSB_EVENT_ERR_WOR</i>	<i>Write FIFO over-run error</i>

Remark

None

4.5 GPSB State

Constant values for the GPSB transfer state management

Declaration

```
#define GPSB_XFER_STATE_DISABLED    (uint8)(0x00UL)
#define GPSB_XFER_STATE_IDLE       (uint8)(0x01UL)
#define GPSB_XFER_STATE_RUNNING    (uint8)(0x02UL)
```

Description

Value	Description
<i>GPSB_XFER_STATE_DISABLED</i>	<i>GPSB is disabled.</i>
<i>GPSB_XFER_STATE_IDLE</i>	<i>GPSB is in idle state.</i>
<i>GPSB_XFER_STATE_RUNNING</i>	<i>GPSB is in running state. Other transfer requests are blocked in this state.</i>

Remark

None

5 STRUCTURES

5.1 GPSBDev_t

Structure to store operating information of each GPSB channel. GPSBDev_t is declared in *gpsb_reg.c* and has a default initial value. If you input parameter values when calling API such as *GPSB_Open*, the parameter values are automatically stored in this struct member variable.

Declaration

```
typedef struct GPSBDev
{
    Uint32          dChannel;
    uint32          dBase;
    uint32          dCfgAddr;
    uint32          dPeriName;
    uint32          dIobusName;
    uint32          dIsSlave;
    uint32          dSdo;
    uint32          dSdi;
    uint32          dSclk;
    uint32          dPort;
    uint32          dIntrPriority;
    uint32          dSupportDma;
    GPSBDmaBuff_t   dTxDma;
    GPSBDmaBuff_t   dRxDma;
    const void *     dAsyncTxBuf;
    void *           dAsyncRxBuf;
    uint32           dAsyncTxDataSize;
    uint32           dAsyncRxDataSize;
    uint8            dState;
    GPSBXferComplite_t dComplete;
} GPSBDev_t;
```

Description

Value	Description
<i>dChannel</i>	<i>GPSB channel number</i>
<i>dBase</i>	<i>GPSB channel base address</i>
<i>dCfgAddr</i>	<i>GPSB port configuration address</i>
<i>dPeriName</i>	<i>Peri clock index</i>
<i>dIobusName</i>	<i>Peripherals IOBUS index</i>
<i>dIsSlave</i>	<i>GPSB mode (Master or Slave)</i>
<i>dSdo</i>	<i>SDO (MOSI) GPIO index information</i>
<i>dSdi</i>	<i>SDI (MISO) GPIO index information</i>

<i>Value</i>	<i>Description</i>
<i>dSclk</i>	<i>SCLK GPIO index information</i>
<i>dPort</i>	<i>Matched port number</i>
<i>dIntrPriority</i>	<i>Interrupt priority</i>
<i>dSupportDma</i>	<i>Support information</i> <i>Default: FALSE</i>
<i>dTxDma</i>	<i>Tx DMA Buffer information</i>
<i>dRxDma</i>	<i>Rx DMA Buffer information</i>
<i>dAsyncTxBuf</i>	<i>Tx buffer address for FIFO async transfer</i>
<i>dAsyncRxBuf</i>	<i>Rx buffer address for FIFO async transfer</i>
<i>dAsyncTxDataSize</i>	<i>Async FIFO transfer Tx data size</i>
<i>dAsyncRxDataSize</i>	<i>Async FIFO transfer Rx data size</i>
<i>dState</i>	<i>State of GPSB</i>
<i>dComplete</i>	<i>Notify the completion of non-blocking transfer</i>

Remark

None

5.2 GPSBXferComplete_t

Structure to register callback function. The value of *xcCallback* parameters is filled when calling *GPSB_Open*.

Declaration

```
typedef struct GPSBXferComplete
{
    GPSBCallback    xcCallback;
    void *          xcArg;
} GPSBXferComplite_t;
```

Description

Value	Description
<i>xcCallback</i>	Callback function pointer. Refer to GPSBCallback .
<i>xcArg</i>	User parameter

Remark

None

6 CALLBACK FUNCTIONS

6.1 GPSBCallback

When async transfer is completed or an error interrupt occurs, the GPSB driver notifies you of the event information via a callback function.

The supported events are as follows:

- Transmit done.
- Error interrupts.

Declaration

```
typedef void (*GPSBCallback)
(
    uint32 uiCh,
    int32 iEvent,
    void * pArg
);
```

Parameters

Value	Description
<i>uiCh</i>	[in] Value of channel to be controlled
<i>iEvent</i>	[in] Value of Event information. Refer to GPSB Event .
<i>pArg</i>	[in] User parameter

Return

None

Remark

None

7 APIs

7.1 GPSB_Init

Registers the GPSB interrupt handler functions and enables the GPSB interrupts. Normally, interrupts occur when using FIFO or DMA async transfer.

The supported interrupts: GPSB 0 ~ 4, GPSB DMA 0 ~ 4

Declaration

```
void GPSB_Init  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7.2 GPSB_Deinit

Disables all the GPSB interrupts.

Declaration

```
void GPSB_Deinit  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7.3 GPSB_Reset

Resets the GPSB channel in Peripherals IOBUS. After reset, the GPSB channel register is set to default value.

Declaration

```
SALRetCode_t GPSB_Reset
(
    Uint32 uiCh
);
```

Parameters

Value	Description
uiCh	[in] Value of channel to be controlled

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.4 GPSB_Open

Initializes the GPSB channel and inputs valid parameter to operate the GPSB channel. SCK, SDO, and SDI values are different depending on HW construction (When using an internal path such as loopback mode, SDO/SDI/SCK can be set to 0). If want to get notification from the GPSB driver, register a callback function.

Declaration

```
SALRetCode_t GPSB_Open
(
    uint32      uiCh,
    uint32      uiSdo,
    uint32      uiSdi,
    uint32      uiSclk,
    void *      pDmaAddrTx,
    void *      pDmaAddrRx,
    uint32      uiDmaBufSize,
    GPSBCallback fbCallback,
    void *      pArg
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>[in] Value of channel to be controlled</i>
<i>uiSdo</i>	<i>[in] SDO GPIO index</i>
<i>uiSdi</i>	<i>[in] SDI GPIO index</i>
<i>uiSclk</i>	<i>[in] SCLK GPIO index</i>
<i>pDmaAddrTx</i>	<i>[in] DMA TX buffer physical address</i>
<i>pDmaAddrRx</i>	<i>[in] DMA RX buffer physical address</i>
<i>uiDmaBufSize</i>	<i>[in] DMA buffer size</i>
<i>fbCallback</i>	<i>[in] Callback function pointer. Refer to the GPSBCallback.</i>
<i>pArg</i>	<i>[in] User parameter</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.5 GPSB_Close

De-initializes GPSB channel.

- Stop DMA.
- Disable channel interrupt.
- Clear register value to disable GPSB channel.

Declaration

```
SALRetCode_t GPSB_Close  
(  
    uint32 uiCh  
);
```

Parameters

Value	Description
uiCh	[in] Value of channel to be controlled

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.6 GPSB_SetSpeed

Sets GPSB operating clock speed. The maximum operating clock speed is limited by HW construction and signal strength.

Declaration

```
SALRetCode_t GPSB_SetSpeed  
(  
    uint32 uiCh,  
    uint32 uiSpeedInHz  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>[in] Value of channel to be controlled</i>
<i>uiSpeedInHz</i>	<i>[in] SCLK speed</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.7 GPSB_SetBpw

Sets the BPW value. Valid BPW value is 8, 16, or 32.

Declaration

```
SALRetCode_t GPSB_SetBpw  
(  
    uint32 uiCh,  
    uint32 uiBpw  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>[in] Value of channel to be controlled</i>
<i>uiBpw</i>	<i>[in] Bit per width</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.8 GPSB_SetMode

Sets the GPSB operating mode. Polarity and phase of the GPSB signal change depending on the mode value. Refer to GPSB MODE register & Timing Diagram in "*TCC70xx Full Specification*".

Declaration

```
void GPSB_SetMode
(
    uint32 uiCh,
    uint32 uiMode
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>[in] Value of channel to be controlled</i>
<i>uiMode</i>	<i>[in] Operating Mode. (Example: GPSB_CPOL, GPSB_CPHA, GPSB_LOOP, etc.)</i>

Return

None

Remark

None

7.9 GPSB_CsInit

Initializes CS GPIO to default config.

Declaration

```
SALRetCode_t GPSB_CsInit
(
    uint32 uiCh,
    uint32 uiCs,
    uint32 uiCsHigh
);
```

Parameters

Value	Description
uiCh	[in] Value of channel to be controlled
uiCs	[in] CS GPIO index
uiCsHigh	[in] Default high or low

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.10 GPSB_CsAlloc

Configures the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsAlloc  
(  
    uint32 uiCh,  
    uint32 uiCs  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>[in] Value of channel to be controlled</i>
<i>uiCs</i>	<i>[in] CS GPIO index</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.11 GPSB_CsActivate

Activates the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsActivate
(
    uint32 uiCh,
    uint32 uiCs,
    uint32 uiCsHigh
);
```

Parameters

Value	Description
uiCh	[in] Value of channel to be controlled
uiCs	[in] CS GPIO index
uiCsHigh	[in] Default High or Low

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.12 GPSB_CsDeactivate

Deactivates the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsDeactivate  
(  
    uint32 uiCh,  
    uint32 uiCs,  
    uint32 uiCsHigh  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>[in] Value of channel to be controlled</i>
<i>uiCs</i>	<i>[in] CS GPIO index</i>
<i>uiCsHigh</i>	<i>[in] Default High or Low</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.13 GPSB_Xfer

Transmits data. This function can be called by the application. Before calling this function, the GPSB channel must be initialized and configured (speed, BPW, and so on).

Declaration

```
SALRetCode_t GPSB_Xfer
(
    uint32      uiCh,
    const void * pTx,
    void *      pRx,
    uint32      uiDataSize,
    uint32      uiXferMode
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>[in] Value of channel to be controlled</i>
<i>pTx</i>	<i>[in] TX data buffer</i>
<i>pRx</i>	<i>[out] RX data buffer</i>
<i>uiDataSize</i>	<i>[in] Write/Read data size</i>
<i>uiXferMode</i>	<i>[in] Transfer mode (DMA/PIO, Interrupt/Non-Interrupt)</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.14 GPSB_XferAbort

Stops transmission.

Declaration

```
void GPSB_XferAbort
(
    uint32 uiCh
);
```

Parameters

<i>Value</i>	<i>Description</i>
<i>uiCh</i>	<i>[in] Value of channel to be controlled</i>

Return

<i>Value</i>	<i>Description</i>
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

8 HOW TO USE GPSB DRIVER

GPSB driver requires channel initializing & de-initializing step before and after the data transfer.

■ Initialize

Normally, you can set up channels by calling ***GPSB_Open*** with valid parameters. SCK, SDI, SDO, and CS GPIO should be set when using an external path. Operating values such as speed, BPW, and mode can be set after completing ***GPSB_Open*** without error.

■ Data transfer

After initializing, data can be transferred by calling ***GPSB_Xfer***. ***GPSB_Xfer*** requires valid buffer and xfer mode information. Before and after calling ***GPSB_Xfer***, if CTF mode is not used, you should set CS GPIO to control xfer start and stop. Then call ***GPSB_CsActivate*** & ***GPSB_CsDeactivate*** to control the CS GPIO.

■ De-initialize

After GPSB channel use is completed, close GPSB channel by calling ***GPSB_Close*** to prevent unnecessary resource consumption and errors due to redundant use of resources.

8.1 Operation of GPSB

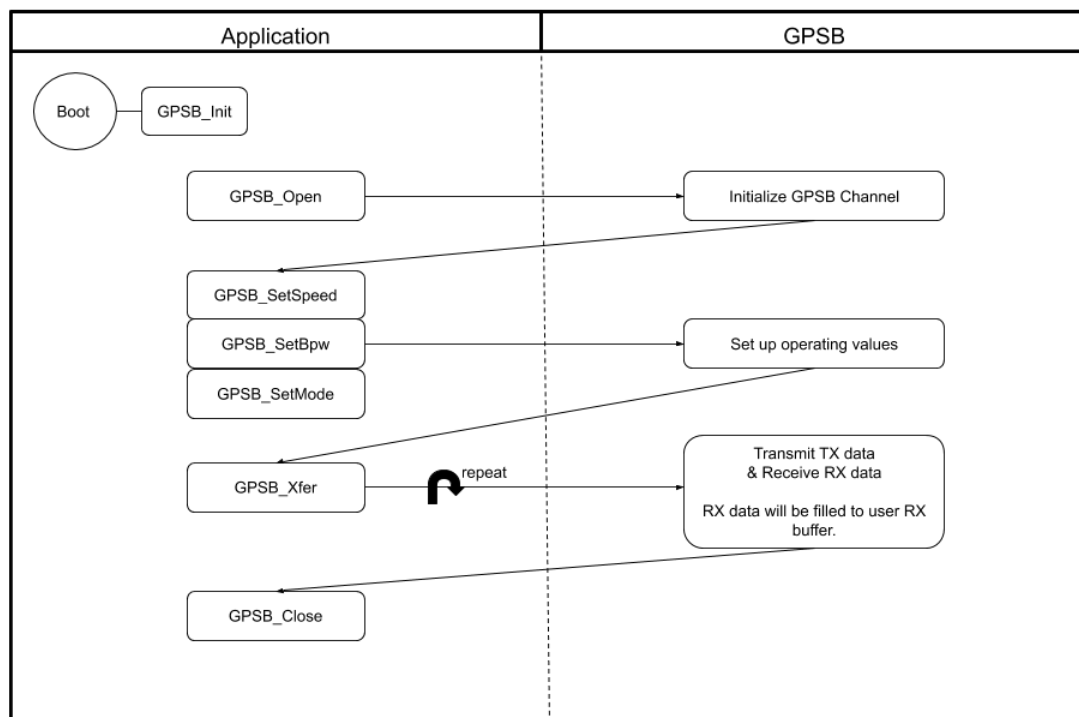


Figure 8.1 Diagram for GPSB Driver

8.2 Example Code

Refer to the test code in *gpsb_test.c* file.

```
static uint32 *   g_tx_dma;
static uint32 *   g_rx_dma;
static uint8      g_tx[GPSB_TEST_BUF_SIZE] = {0,};
static uint8      g_rx[GPSB_TEST_BUF_SIZE] = {0,};

{
    ucCh          = 0UL;
    dma_buf_size   = DMA_BUF_SIZE;
    gpio_sclk      = GPIO_GPB(25UL);
    gpio_cs        = GPIO_GPB(26UL);
    gpio_sdo       = GPIO_GPB(27UL);
    gpio_sdi       = GPIO_GPB(28UL);
    speed          = 20*1000*1000;
    ucBpw         = 16;
    cs_high        = 0;
    xfer_mode      = GPSB_XFER_MODE_WITHOUT_INTERRUPT;

    // initialize channel
    GPSB_Open(ucCh, gpio_sdo, gpio_sdi, gpio_sclk, (void *)g_tx_dma, (void *)g_rx_dma,
              dma_buf_size, NULL, NULL_PTR);

    // configuration
    GPSB_SetSpeed(ucCh, speed);
    GPSB_SetBpw(ucCh, ucBpw);
    GPSB_SetMode(ucCh, 0);

    // Set TX data in buffer (g_tx)
    GPSB_CsActivate(ucCh, gpio_cs, cs_high);
    GPSB_Xfer(ucCh, g_tx, g_rx, data_size, xfer_mode);
    GPSB_CsDeactivate(ucCh, gpio_cs, cs_high);

    // Check RX data in buffer (g_rx)

    // Disable channel
    GPSB_Close(ucCh);
}
```

9 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

10 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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